Cyber Incidents Before and After the Rise of Generative AI

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Project Proposal

Due to the recent news headlines that we have seen on how AI has been used to hack into many different technologies, we want to see how that might look historically, going back to pre-AI cyberattacks. The dataset that we want to use for our project is the European Repository of Cyber Incidents.[1] We are interested in looking at a large database consisting of different cyberattacks that have happened throughout the years all around the world. This data contains information on date, kind of attack, who was targeted, who committed the attack (if known), and lots of other information. Our specific interest lies in looking for relationships that may exist between the proliferation of AI, such as ChatGPT, and the number, kinds, and frequency of attacks that have happened throughout the years.

Our overall research question is: How have cyberattacks changed alongside the rise of widely accessible AI technologies? This allows us to examine whether the frequency, types, impacts, targets, and attribution patterns of recorded cyber incidents have changed over time.

In essence, our working hypothesis is that the frequency of cyberattacks has increased as AI technologies have become more accessible, due to the access to information that can be misused through AI chatbots. We want to focus on the years 2017 to 2023, using 2020 as the dividing point between the two periods.[2] We will also be looking at specific regions that receive more cyberattacks, the types of attacks being launched, and whether the attackers are known or unknown. We also believe that the amount of unknown or "no-name" attackers may have increased during this period.

In addition to these expectations, we will examine whether the impacts of cyber incidents have changed between the two periods, including changes in cyber intensity, functional impact, disruption, and data theft.

More specifically, for the data we will be using, we have downloaded a CSV file from the European Repository of Cyber Incidents. The dataset contains approximately 4,800 recorded

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[1] European Repository of Cyber Incidents: https://eurepoc.eu/table-view/

[2] Though AI existed before 2020, we use 2020 as our dividing point because GPT-3 was introduced that year and access to generative AI models began to expand. This marked an important shift toward generative AI becoming more accessible and widely used. *Nature Machine Intelligence*, “Next chapter in artificial writing” (2020): https://www.nature.com/articles/s42256-020-0223-0

cyber incidents and 84 variables, with each row representing one recorded cyber incident. The European Repository of Cyber Incidents collects information from publicly reported cyber incidents and uses a combination of data science tools and analysis to code and maintain its database. This is a lot of information that we do not need, so we will be taking steps to clean up the data. First, we will cut the data down to include the columns that are relevant to our research questions. These will include date and status variables such as `start_date`, `end_date`, and `status`; incident type and impact variables such as `incident_type` `weighted_cyber_intensity`, `has_disruption`, `data_theft`, and `functional_impact`; and target and attribution variables such as receiver_country, `receiver_category`, `initiator_country`, `initiator_category`, and `number_of_attributions`.

We will also clean and process the data by converting dates into a consistent format and creating a year variable to examine changes over time. We will examine missing and unknown values, determine how to handle incidents with an `Open` status, and create a variable separating incidents into our two comparison periods. We will also decide how to categorize incidents as known or `unknown` and `attributed` or `unattributed`.
The research questions we want to answer are:

1. How has the frequency of recorded cyber incidents changed before and after our 2020 dividing point?
2. What types of attacks were used in the past and now? More specifically, have the types and impacts of recorded cyber incidents changed between the two periods?
3. Who was being attacked in the past and now, who were the attackers, and what regions were being attacked? More specifically, have the targets and known/unknown initiators of recorded cyber incidents changed over this period?

To answer these questions, we plan to compare frequencies, percentages, distributions, and means or medians between the two periods. We will examine the number of recorded cyber incidents per year, changes in attack types, average or median cyber intensity, the percentage of incidents involving disruption or data theft, the most frequently targeted sectors and regions, and the proportion of incidents involving known, unknown, attributed, or unattributed initiators. We will use tables and visualizations to examine these patterns and how they change over time. One limitation of this analysis is that the dataset does not indicate whether AI was actually used to carry out an individual cyberattack. Therefore, our analysis can identify changes in recorded

cyber incidents that occurred alongside the rise of widely accessible AI, but it cannot establish that AI directly caused those changes.

Group Contract

For our division of labor, we will plan the project in different parts and divide the work within each part among the three group members. Some tasks may depend on another group member completing their work first, so we will communicate these dependencies and set deadlines to make sure the project stays on track. We will divide the workload as evenly as possible, while also helping each other when needed. Everyone will help with preparing the data, analyzing it, writing the report, and creating the final presentation. We will divide up specific tasks as we work through each part of the project.

We will communicate regularly about our progress and make sure everyone knows what they are responsible for and when their work needs to be completed. If someone is having difficulty completing their assigned task or expects a delay, they will communicate this to the group so we can adjust our plan if necessary. Once individual parts are completed, we will review the work together to make sure the final project is consistent and complete.